

are the key expectations or technical drivers for any enterprise to take the cloud route. With the application estate and the infrastructure costs getting higher and higher, it is always a complex task to decide how to optimise cloud consumption and cost. Cloud service providers like Azure, AWS or GCP provide advisory services, which helps to optimise their services and usage. This may be useful in the case of native services but when it comes to the IaaS or PaaS models, we may need something more cutting-edge than that.

How to manage cloud cost

We need to focus on the following points to ensure that cloud cost management is more effective.

- **Understand pricing models:** Each cloud vendor has a different pricing model for different services. If you are a public cloud user, be sure to look into the fine print on what the pricing model entails.
- **Identify key cloud cost contributors:** Based on your cloud spend, identify services that are key contributors to your monthly spend, and focus on them with a top-to-bottom approach.
- **Analyse workload usage patterns:** Detect usage trends for each of the services utilised on the public cloud, paying special attention to the key cloud cost contributors. Then, collect metrics like CPU, memory, disk/storage, API, etc.
- **Identify unused resources:** Since every cloud service is available at the click of a button, each one can be provisioned and completely forgotten, leading to orphaned or unused resources that cost thousands of dollars.
- **Excavate archaic data:** Along with the key cloud cost contributors, it's also important to analyse services that have a higher chance of storing purposeless archaic data.
- **Scale down underutilised resources:** Based on your usage patterns, slowly scale down resources to the next smallest size, ensuring the size-down doesn't impact application performance. Continue this exercise until your workload runs at its desired level.
- **Upgrade infrastructure with the latest resources:** Public cloud providers keep optimising their services to ensure security, performance and cost are competitive. So, it's wise to also optimise your infrastructure with those upgrades to stay on top of the game.
- **Leverage on-demand services:** Plan to leverage services on-demand for workloads that don't run in-production. Since development teams can't work 24/7, it's ideal to automate your start/stop workloads.
- **Consolidate accounts:** When running services from public cloud providers on multiple accounts, try to consolidate your accounts—most providers have a volume-based discounting pricing model that can help you optimise cost.
- **Baseline your infrastructure:** While going through the analyse and optimise exercises, it's possible to reach the baseline of your infrastructure, which you need in order to run any workload. There might also be times when your infrastructure needs to be scaled out; so keep those needs in mind.
- **Separate the workloads:** Based on the usage patterns and purposes of your workloads, separate them into different categories like stable, variable, long-term and short-term.
- **Reserve resources:** Once you've identified stable workloads for each environment, reserve these resources. Reservations are smarter financially if the capacity meets different vendors' pricing models. If you have workloads running across multiple accounts on demand in different time zones, consider

reserving some capacity for dynamic workloads.

- **Create business units:** Each workload has a specific function. Based on your organisation's structure, create business units (BUs) and define the owners. Assign these workloads to these business units, ensuring they are responsible for the charge
- **Identify requirements of each BU:** Each BU should provide capacity and scale requirements for their workloads that are stable for the short-term, but flexible enough to meet budgeting demands.
- **Create a budget and chargeback policy:** Based on your requirements, define a budget for each BU. Communicate this budget to each BU owner, making sure they are responsible for staying optimised and within budget.
- **Define cloud cost governance policies:** Policies vary by organisation, but just as reviewing cost summaries can lead to immediate savings, your infrastructure should be provisioned only through automation, ensuring key tags are always assigned to each resource (since not everyone has access to infrastructure provisioning).
- **Go ahead with tagging:** Public cloud providers support tagging infrastructure, which can help identify workloads for management, segregation and billing purposes. Your cloud management platform can then consume these tags and generate different reports.

Models for cloud cost management

Cost optimisation on the cloud platform can be achieved using any of the three models listed below.

DevOps driven model: This is the low hanging fruit with low maturity. It provides a transparent facility for cost visibility and management with a